

Question Bank — Computer Networks

Module 1 companion | PYQ-style + Practice | exam-tagged, answers & explanations

■ How to use this bank

Two sections: PYQ-style = questions modelled on what govt CS/IT exams actually ask (IBPS, SSC, GATE, PSU, DSSSB), tagged by exam & level. **Practice** = full syllabus-coverage set for testing & revision, organised by sub-topic.

Every question has the correct option + explanation in the green box. Attempt first, then check.

Tag legend: IBPS SO IT • Coal India MT • DSSSB TGT • All 3 exams • GATE-level (deeper, mainly Coal India) • SSC/other PSU.

For IBPS (Aug 29): focus on IBPS / All-3 tags; skip GATE-tagged numericals unless targeting Coal India.

Note: these are original questions faithfully modelled on real exam patterns & difficulty — not verbatim reproductions of copyrighted papers.

SECTION A — PYQ-style (modelled on actual exam questions)

Tagged by the exam whose style/level each reflects. ~50 questions.

Basics, models & layers

Q1

All 3 exams

Which OSI layer is responsible for end-to-end error-free delivery and flow control?

- (a) Network
- (b) Transport
- (c) Session
- (d) Data Link

Answer: (b)

Transport layer (L4) provides end-to-end (process-to-process) reliable delivery, flow and congestion control; TCP lives here.

Q2

IBPS SO IT

All 3 exams

The number of layers in TCP/IP model and OSI model respectively are:

- (a) 4 and 7
- (b) 7 and 4
- (c) 5 and 7
- (d) 4 and 5

Answer: (a)

TCP/IP = 4 layers, OSI = 7 layers. TCP/IP merges OSI's session+presentation+application into one Application layer.

Q3

SSC/other

All 3 exams

Which layer of OSI is concerned with syntax and semantics of information exchanged (translation, encryption)?

- (a) Session
- (b) Presentation
- (c) Application
- (d) Transport

Answer: (b)

Presentation layer (L6) handles data translation, encryption/decryption, and compression.

Q4

All 3 exams

In which layer does a repeater operate?

- (a) Physical
- (b) Data Link
- (c) Network
- (d) Transport

Answer: (a)

A repeater regenerates signals at the Physical layer (L1) to extend distance.

Q5

GATE-level

Coal India MT

The header size of a standard IPv4 packet (without options) is:

- (a) 20 bytes
- (b) 40 bytes
- (c) 8 bytes
- (d) 16 bytes

Answer: (a)

IPv4 header is 20 bytes minimum (without options). IPv6 fixed header is 40 bytes; UDP header 8 bytes. GATE-depth detail.

Physical layer & media

Q6

DSSSB TGT

All 3 exams

Which connector is used with twisted-pair Ethernet cables?

- (a) BNC
- (b) RJ45
- (c) SC
- (d) RJ11

Answer: (b)

RJ45 connectors are used with twisted-pair (Cat5e/6) Ethernet. RJ11 is for telephone lines; BNC for coax.

Q7

All 3 exams

Which multiplexing technique assigns different frequency bands to different signals?

- (a) TDM
- (b) FDM
- (c) CDM
- (d) PDM

Answer: (b)

FDM (Frequency Division Multiplexing) gives each signal a distinct frequency band; TDM divides by time slots.

Q8

GATE-level

Coal India MT

By Nyquist's theorem, the max data rate of a noiseless 4 kHz channel carrying 4-level signals is:

- (a) 8 kbps
- (b) 16 kbps
- (c) 32 kbps
- (d) 4 kbps

Answer: (b)

Nyquist: $C = 2 \cdot B \cdot \log_2(L) = 2 \times 4000 \times \log_2(4) = 2 \times 4000 \times 2 = 16000 \text{ bps} = 16 \text{ kbps}$. GATE/Coal-India depth.

Data link, MAC & devices

Q9

All 3 exams

How many bits are there in a MAC address, and how is it usually written?

- (a) 32 bits, 4 octets
- (b) 48 bits, 12 hex digits
- (c) 64 bits, 8 groups
- (d) 128 bits, hex

Answer: (b)

MAC = 48 bits, shown as 12 hexadecimal digits (6 pairs). IPv4=32-bit, IPv6=128-bit.

Q10

All 3 exams

SSC/other

A switch and a bridge both operate at which OSI layer?

- (a) Physical
- (b) Data Link
- (c) Network
- (d) Transport

Answer: (b)

Both switch and bridge are Data Link layer (L2) devices that forward by MAC address.

Q11

All 3 exams

Which device breaks up broadcast domains?

- (a) Hub
- (b) Switch
- (c) Router
- (d) Repeater

Answer: (c)

A router (L3) separates broadcast domains. A switch separates collision domains but forwards broadcasts within a LAN.

Q12

DSSSB TGT

All 3 exams

CSMA/CD is the access-control method used in:

- (a) Wi-Fi
- (b) Traditional Ethernet
- (c) Token Ring
- (d) Bluetooth

Answer: (b)

CSMA/CD (Collision Detection) is classic wired Ethernet. Wi-Fi uses CSMA/CA.

Q13

GATE-level

Coal India MT

In pure ALOHA, the maximum channel utilization (efficiency) is approximately:

- (a) 18.4%
- (b) 36.8%
- (c) 50%
- (d) 100%

Answer: (a)

Pure ALOHA max efficiency $\approx 18.4\%$ ($1/2e$); slotted ALOHA $\approx 36.8\%$ ($1/e$). GATE-depth MAC theory.

Q14

All 3 exams

Which error-detection scheme can also correct single-bit errors?

- (a) CRC
- (b) Checksum
- (c) Parity
- (d) Hamming code

Answer: (d)

Hamming code adds redundant parity bits at power-of-two positions to detect and correct single-bit errors; CRC/checksum/parity only detect.

Network layer, IP & subnetting

Q15

All 3 exams

Which IP address class provides the largest number of hosts per network?

- (a) Class A
- (b) Class B
- (c) Class C
- (d) Class D

Answer: (a)

Class A (/8) has 24 host bits → the most hosts per network. Class D is multicast (no host concept).

Q16

IBPS SO IT

All 3 exams

The default subnet mask for a Class B address is:

- (a) 255.0.0.0
- (b) 255.255.0.0
- (c) 255.255.255.0
- (d) 255.255.255.255

Answer: (b)

Class B default mask = 255.255.0.0 (/16). Class A = /8, Class C = /24.

Q17

IBPS SO IT

GATE-level

How many subnets are created by borrowing 3 bits from the host portion?

- (a) 3
- (b) 6
- (c) 8
- (d) 9

Answer: (c)

Subnets = $2^b = 2^3 = 8$ (classless counting). Number of borrowed bits directly gives the power of two.

Q18

IBPS SO IT

GATE-level

For the network 192.168.1.0/28, how many usable hosts per subnet?

- (a) 16
- (b) 14
- (c) 30
- (d) 62

Answer: (b)

/28 → $32 - 28 = 4$ host bits → $2^4 = 16$, minus 2 = 14 usable hosts.

Q19

All 3 exams

The loopback address in IPv4 is:

- (a) 0.0.0.0
- (b) 127.0.0.1
- (c) 192.168.0.1
- (d) 255.255.255.255

Answer: (b)

127.0.0.1 is the loopback (localhost) address used to test the local TCP/IP stack.

Q20

All 3 exams

SSC/other

Which protocol maps an IP address to a physical (MAC) address?

- (a) RARP
- (b) ARP
- (c) DHCP
- (d) DNS

Answer: (b)

ARP resolves IP → MAC on the local link. RARP does the reverse (MAC → IP).

Q21

All 3 exams

NAT is primarily used to:

- (a) Encrypt packets
- (b) Map private IPs to public IPs
- (c) Assign MAC addresses
- (d) Compress data

Answer: (b)

NAT (Network Address Translation) maps many private IPs to one/few public IPs, easing IPv4 exhaustion and adding a layer of hiding.

Q22

Coal India MT

GATE-level

OSPF is best classified as which type of routing protocol?

- (a) Distance vector
- (b) Link state
- (c) Path vector
- (d) Static

Answer: (b)

OSPF is a link-state protocol using Dijkstra's algorithm. RIP is distance-vector; BGP is path-vector.

Q23

GATE-level

Coal India MT

The maximum number of hops allowed in RIP before a route is considered unreachable is:

- (a) 15
- (b) 16
- (c) 31
- (d) 255

Answer: (a)

RIP treats 16 as infinity (unreachable), so 15 is the max usable hop count — limiting RIP to small networks.

Q24

All 3 exams

Which field in the IPv4 header prevents packets from looping forever?

- (a) Checksum
- (b) TTL
- (c) Flags
- (d) Protocol

Answer: (b)

TTL (Time To Live) is decremented by each router; at zero the packet is discarded, preventing infinite loops.

Transport layer

Q25

All 3 exams

Which is a connectionless transport protocol?

- (a) TCP
- (b) UDP
- (c) IP
- (d) ARP

Answer: (b)

UDP is connectionless and unreliable but fast; TCP is connection-oriented and reliable.

Q26

All 3 exams

SSC/other

The TCP header size (without options) is:

- (a) 8 bytes
- (b) 20 bytes
- (c) 16 bytes
- (d) 40 bytes

Answer: (b)

TCP header = 20 bytes minimum (like IPv4). UDP header is only 8 bytes.

Q27

IBPS SO IT

All 3 exams

Which flags are exchanged during the TCP three-way handshake?

- (a) SYN, SYN-ACK, ACK
- (b) FIN, ACK, RST
- (c) SYN, FIN, ACK
- (d) ACK, PSH, URG

Answer: (a)

Handshake: SYN → SYN-ACK → ACK. FIN/RST are for teardown/reset.

Q28

GATE-level

Coal India MT

In TCP congestion control, the phase where the window grows exponentially is:

- (a) Congestion avoidance
- (b) Slow start
- (c) Fast recovery
- (d) Fast retransmit

Answer: (b)

Slow start grows the congestion window exponentially until the threshold; then congestion avoidance grows it linearly. GATE-depth.

Q29

All 3 exams

Port numbers are how many bits, giving what range?

- (a) 8 bits, 0-255
- (b) 16 bits, 0-65535
- (c) 32 bits
- (d) 10 bits, 0-1023

Answer: (b)

Port numbers are 16-bit → range 0-65535. Well-known ports are 0-1023.

Application layer & ports

Q30

All 3 exams

DSSSB TGT

Which protocol and port does HTTPS use?

- (a) TCP 80
- (b) TCP 443
- (c) UDP 53
- (d) TCP 25

Answer: (b)

HTTPS = HTTP over TLS on TCP port 443. HTTP=80, DNS=UDP 53, SMTP=25.

Q31

All 3 exams

DNS mainly uses which transport protocol and port for standard queries?

- (a) TCP 53
- (b) UDP 53
- (c) TCP 25
- (d) UDP 67

Answer: (b)

DNS uses UDP port 53 for normal queries (TCP 53 for zone transfers / large responses).

Q32

All 3 exams

SSC/other

Which protocol is used to retrieve email and keeps messages synchronized across devices?

- (a) SMTP
- (b) POP3
- (c) IMAP
- (d) FTP

Answer: (c)

IMAP (143) syncs mail across devices; POP3 (110) downloads and typically removes from server; SMTP (25) sends.

Q33

IBPS SO IT

All 3 exams

FTP uses which ports for control and data respectively?

- (a) 20 and 21
- (b) 21 and 20
- (c) 22 and 23
- (d) 80 and 443

Answer: (b)

FTP uses port 21 for control and port 20 for data transfer.

Q34**DSSSB TGT****All 3 exams**

In a URL, the component resolved by DNS is the:

- (a) Path
- (b) Protocol
- (c) Domain name
- (d) Query string

Answer: (c)

DNS resolves the domain name to an IP address before the HTTP request is made.

IPv6, security & misc

Q35**All 3 exams**

IPv6 uses addresses of length:

- (a) 32 bits
- (b) 64 bits
- (c) 128 bits
- (d) 256 bits

Answer: (c)

IPv6 = 128-bit addresses, written as eight groups of hex, solving IPv4 exhaustion.

Q36**IBPS SO IT****All 3 exams**

A firewall that inspects the state of active connections is called:

- (a) Packet filter
- (b) Stateful firewall
- (c) Proxy
- (d) IDS

Answer: (b)

A stateful firewall tracks connection state (not just individual packets), allowing return traffic for established sessions.

Q37**IBPS SO IT****DSSSB TGT**

A VPN primarily provides:

- (a) More bandwidth
- (b) An encrypted tunnel over a public network
- (c) IP assignment
- (d) Name resolution

Answer: (b)

A VPN creates a secure encrypted tunnel across a public network for private, remote access.

Q38**All 3 exams**

A DDoS attack primarily targets which security goal?

- (a) Confidentiality
- (b) Integrity
- (c) Availability
- (d) Non-repudiation

Answer: (c)

DoS/DDoS floods a system to deny service, attacking Availability in the CIA triad.

Q39

Coal India MT

GATE-level

Which switching creates a dedicated path for the entire duration of a session?

- (a) Packet switching
- (b) Datagram switching
- (c) Circuit switching
- (d) Message switching

Answer: (c)

Circuit switching reserves an end-to-end path for the whole session (telephone model).

Q40

All 3 exams

SSC/other

Which topology requires the most cabling for full connectivity?

- (a) Bus
- (b) Star
- (c) Ring
- (d) Mesh

Answer: (d)

A full mesh needs $n(n-1)/2$ links — the most cabling, but the most fault-tolerant.

Q41

DSSSB TGT

All 3 exams

WWW, email and file transfer are examples of services at which layer?

- (a) Transport
- (b) Network
- (c) Application
- (d) Session

Answer: (c)

These are Application-layer services (HTTP, SMTP, FTP).

Q42

All 3 exams

Which device amplifies/regenerates a weak signal to extend network distance?

- (a) Router
- (b) Repeater
- (c) Bridge
- (d) Gateway

Answer: (b)

A repeater (L1) regenerates the signal to cover longer distances.

Q43

GATE-level

Coal India MT

In Selective Repeat ARQ with sequence number field of m bits, the max window size is:

- (a) 2^m
- (b) $2^m - 1$
- (c) $2^{(m-1)}$
- (d) m

Answer: (c)

Selective Repeat max window = $2^{(m-1)}$. Go-Back-N allows $2^m - 1$. GATE-depth.

Q44

IBPS SO IT

All 3 exams

Which of these is NOT a private IP range?

- (a) 10.0.0.0/8
- (b) 172.16.0.0/12
- (c) 192.168.0.0/16
- (d) 172.32.0.0/12

Answer: (d)

Private Class B range is 172.16.0.0 – 172.31.255.255. 172.32.x.x is public, so it is NOT private.

Q45

All 3 exams

The process of adding headers as data moves down the stack is:

- (a) Decapsulation
- (b) Encapsulation
- (c) Fragmentation
- (d) Segmentation

Answer: (b)

Encapsulation adds each layer's header going down; decapsulation removes them going up.

Q46

All 3 exams

SSC/other

Bluetooth is an example of which network type?

- (a) LAN
- (b) WAN
- (c) PAN
- (d) MAN

Answer: (c)

Bluetooth forms a PAN (Personal Area Network) — short-range, around a person.

Q47

DSSSB TGT

All 3 exams

Which of the following is an example of guided (wired) media?

- (a) Radio waves
- (b) Microwaves
- (c) Optical fibre
- (d) Infrared

Answer: (c)

Optical fibre is guided (wired) media. Radio/microwave/infrared are unguided (wireless).

SECTION B — Practice (full syllabus coverage for revision)

Organised by sub-topic. Attempt as timed sets or topic-wise revision.

B1. Foundations, types & topologies

Q48

All 3 exams

A network spanning a city, typically run by an ISP or cable operator, is a:

- (a) LAN
- (b) MAN
- (c) WAN
- (d) PAN

Answer: (b)

MAN (Metropolitan Area Network) covers a city-sized area.

Q49

All 3 exams

Full-duplex communication means:

- (a) One direction only
- (b) Both directions, one at a time
- (c) Both directions simultaneously
- (d) No direction

Answer: (c)

Full-duplex = both directions at once (e.g. telephone). Half-duplex = one at a time; simplex = one way only.

Q50

All 3 exams

In a ring topology, data typically travels:

- (a) To all nodes at once
- (b) In one direction node to node
- (c) Only to the server
- (d) Randomly

Answer: (b)

In a ring, each node passes data to its neighbour, circulating around the ring (often token-controlled).

Q51

All 3 exams

Which topology is easiest to troubleshoot because each node has its own link to a central device?

- (a) Bus
- (b) Ring
- (c) Star
- (d) Mesh

Answer: (c)

Star isolates node faults to individual links, easing troubleshooting.

Q52

All 3 exams

A hybrid topology is:

- (a) Only bus + ring
- (b) A combination of two or more basic topologies
- (c) Same as mesh
- (d) A wireless-only topology

Answer: (b)

A hybrid combines two or more basic topologies (e.g. star-bus).

Q53

SSC/other

All 3 exams

Which topology continues to work if one node fails but not if the backbone cable breaks?

- (a) Star
- (b) Mesh
- (c) Bus
- (d) Ring

Answer: (c)

In a bus, a node failure is tolerated but a backbone break disables the whole segment.

Q54

All 3 exams

The number of I/O ports each device needs in a full mesh of n nodes is:

- (a) n
- (b) $n-1$
- (c) $n(n-1)/2$
- (d) $2n$

Answer: (b)

Each node connects to the other $(n-1)$ nodes, needing $(n-1)$ ports.

B2. Transmission media & switching

Q55

All 3 exams

Which cable type has a central conductor surrounded by insulation and a braided shield?

- (a) Twisted pair
- (b) Coaxial
- (c) Fibre
- (d) STP

Answer: (b)

Coaxial cable has a central conductor, dielectric insulation, and a braided metallic shield.

Q56

All 3 exams

Which media offers the greatest resistance to eavesdropping and signal tapping?

- (a) Twisted pair
- (b) Coaxial
- (c) Optical fibre
- (d) Radio

Answer: (c)

Fibre carries light and is very hard to tap without detection; it's also immune to EMI.

Q57

All 3 exams

Microwave transmission requires:

- (a) Underground cables
- (b) Line-of-sight between antennas
- (c) Twisted pair
- (d) Fibre backbone

Answer: (b)

Microwaves are directional and need line-of-sight between transmitting/receiving antennas.

Q58

All 3 exams

In packet switching, packets of the same message may:

- (a) Always take the same path
- (b) Take different paths and arrive out of order
- (c) Never be reordered
- (d) Use a reserved circuit

Answer: (b)

In datagram packet switching, packets are routed independently and may arrive out of order.

Q59

Coal India MT

All 3 exams

Virtual circuit switching is:

- (a) Connectionless
- (b) Connection-oriented with a preset path
- (c) Same as datagram
- (d) Used by IP

Answer: (b)

A virtual circuit sets up a logical path first; all packets follow it in order (connection-oriented), e.g. MPLS/ATM.

Q60

All 3 exams

TDM (Time Division Multiplexing) shares a link by:

- (a) Frequency bands
- (b) Time slots
- (c) Codes
- (d) Wavelengths

Answer: (b)

TDM allocates the channel in time slots to different signals.

B3. OSI & TCP/IP models

Q61

All 3 exams

Which OSI layer establishes, manages and terminates sessions between applications?

- (a) Transport
- (b) Session
- (c) Presentation
- (d) Application

Answer: (b)

Session layer (L5) manages dialogues/sessions between applications.

Q62

All 3 exams

The PDU at the Network layer is a:

- (a) Frame
- (b) Segment
- (c) Packet
- (d) Bit

Answer: (c)

Network layer PDU = packet. Transport=segment, Data Link=frame, Physical=bit.

Q63

All 3 exams

Which layer converts frames into bits for transmission on the medium?

- (a) Data Link
- (b) Physical
- (c) Network
- (d) Transport

Answer: (b)

The Physical layer transmits raw bits over the medium.

Q64

SSC/other

All 3 exams

In TCP/IP, the Internet layer corresponds to which OSI layer?

- (a) Transport
- (b) Network
- (c) Data Link
- (d) Session

Answer: (b)

TCP/IP Internet layer $\equiv$ OSI Network layer (IP, ICMP, ARP).

Q65

All 3 exams

Which of the following is a Transport-layer protocol?

- (a) IP
- (b) ICMP
- (c) UDP
- (d) ARP

Answer: (c)

UDP (and TCP) are Transport-layer protocols; IP/ICMP/ARP are at the Internet/Network level.

Q66

All 3 exams

Which OSI layer does data compression belong to?

- (a) Application
- (b) Presentation
- (c) Session
- (d) Transport

Answer: (b)

Presentation layer handles compression, translation, and encryption.

Q67

All 3 exams

The mnemonic 'All People Seem To Need Data Processing' represents OSI layers from:

- (a) Bottom to top
- (b) Top to bottom
- (c) Randomly
- (d) Network outward

Answer: (b)

It maps top-to-bottom: Application, Presentation, Session, Transport, Network, Data Link, Physical.

B4. Data link layer & MAC

Q68

All 3 exams

The two sublayers of the Data Link layer are:

- (a) MAC and LLC
- (b) IP and TCP
- (c) ARP and RARP
- (d) DNS and DHCP

Answer: (a)

Data Link splits into LLC (Logical Link Control) and MAC (Media Access Control).

Q69

All 3 exams

Framing is a function of which layer?

- (a) Physical
- (b) Data Link
- (c) Network
- (d) Transport

Answer: (b)

Framing (delimiting bits into frames) is a Data Link layer job.

Q70

All 3 exams

CSMA/CA is used primarily in:

- (a) Ethernet
- (b) Wi-Fi (wireless LAN)
- (c) Token Ring
- (d) FDDI

Answer: (b)

CSMA/CA (Collision Avoidance) is used by Wi-Fi, where collisions can't be detected.

Q71

All 3 exams

Parity checking can detect:

- (a) All errors
- (b) Only single-bit errors reliably
- (c) Only burst errors
- (d) No errors

Answer: (b)

A single parity bit detects single-bit (odd-count) errors but misses even-count errors.

Q72

Coal India MT

GATE-level

CRC uses which mathematical operation on the data and generator polynomial?

- (a) Addition
- (b) Binary (modulo-2) division
- (c) Multiplication
- (d) XOR of bytes only

Answer: (b)

CRC divides the data polynomial by a generator polynomial (modulo-2), sending the remainder as the check.

Q73

All 3 exams

In token passing, a station may transmit only when it:

- (a) Detects a collision
- (b) Holds the token
- (c) Senses an idle line
- (d) Receives an ACK

Answer: (b)

Only the token holder may transmit, preventing collisions (Token Ring).

B5. IP addressing & subnetting

Q74

All 3 exams

The first octet range 128-191 corresponds to which class?

- (a) Class A
- (b) Class B
- (c) Class C
- (d) Class D

Answer: (b)

Class B first octet: 128-191. A=0-127, C=192-223, D=224-239.

Q75

IBPS SO IT

GATE-level

How many usable hosts in a /30 subnet?

- (a) 4
- (b) 2
- (c) 6
- (d) 1

Answer: (b)

/30 → 2 host bits → 4 addresses – 2 = 2 usable (common for point-to-point links).

Q76

IBPS SO IT

GATE-level

The subnet mask /25 in dotted decimal is:

- (a) 255.255.255.0
- (b) 255.255.255.128
- (c) 255.255.255.192
- (d) 255.255.254.0

Answer: (b)

/25 = 255.255.255.128 (one bit borrowed into the 4th octet).

Q77

All 3 exams

Which address is the broadcast address of network 192.168.1.0/24?

- (a) 192.168.1.0
- (b) 192.168.1.1
- (c) 192.168.1.255
- (d) 192.168.1.254

Answer: (c)

The all-ones host part is the broadcast: 192.168.1.255 for a /24.

Q78

All 3 exams

APIPA addresses fall in which range?

- (a) 10.x.x.x
- (b) 169.254.x.x
- (c) 172.16.x.x
- (d) 192.168.x.x

Answer: (b)

APIPA (Automatic Private IP Addressing) uses 169.254.0.0/16 when DHCP fails.

Q79

GATE-level

Coal India MT

Given IP 172.16.10.33/27, the network address is:

- (a) 172.16.10.0
- (b) 172.16.10.32
- (c) 172.16.10.33
- (d) 172.16.10.64

Answer: (b)

/27 blocks are size 32; 33 falls in the 32-63 block, so network = 172.16.10.32.

Q80

All 3 exams

Class D addresses are reserved for:

- (a) Private use
- (b) Multicast
- (c) Loopback
- (d) Experimental

Answer: (b)

Class D (224-239) is multicast; Class E (240-255) is experimental.

B6. Routing & support protocols

Q81

All 3 exams

RIP is an example of which routing algorithm?

- (a) Link state
- (b) Distance vector
- (c) Path vector
- (d) Static

Answer: (b)

RIP is a distance-vector protocol using hop count.

Q82

All 3 exams

Which protocol is used between autonomous systems on the Internet?

- (a) RIP
- (b) OSPF
- (c) BGP
- (d) ICMP

Answer: (c)

BGP (Border Gateway Protocol) routes between autonomous systems — the Internet's backbone routing.

Q83

All 3 exams

DHCP uses which ports?

- (a) 53
- (b) 67 and 68
- (c) 80
- (d) 110

Answer: (b)

DHCP uses UDP ports 67 (server) and 68 (client).

Q84

All 3 exams

traceroute/tracert relies on manipulating which IP field?

- (a) Checksum
- (b) TTL
- (c) Flags
- (d) Protocol

Answer: (b)

traceroute sends packets with increasing TTL to reveal each hop via ICMP time-exceeded messages.

Q85

GATE-level

Coal India MT

Dijkstra's algorithm used in link-state routing computes:

- (a) Minimum spanning tree
- (b) Shortest paths from a source
- (c) All-pairs shortest paths
- (d) Maximum flow

Answer: (b)

Dijkstra computes single-source shortest paths (used by OSPF). GATE-depth.

Q86

All 3 exams

Which protocol reports errors such as 'destination unreachable'?

- (a) ARP
- (b) ICMP
- (c) DHCP
- (d) SMTP

Answer: (b)

ICMP carries control/error messages like destination-unreachable and echo (ping).

B7. Transport layer

Q87

All 3 exams

Which provides ordered, reliable byte-stream delivery?

- (a) UDP
- (b) TCP
- (c) IP
- (d) ICMP

Answer: (b)

TCP delivers an ordered, reliable byte stream with acknowledgements and retransmission.

Q88

All 3 exams

UDP is preferred for DNS queries because:

- (a) It is reliable
- (b) It has low overhead and is fast
- (c) It encrypts data
- (d) It guarantees order

Answer: (b)

DNS queries are small and benefit from UDP's low overhead/speed; retransmission is handled by the resolver.

Q89

All 3 exams

The combination of IP address and port number is called a:

- (a) Frame
- (b) Socket
- (c) Segment
- (d) Datagram

Answer: (b)

A socket = IP + port, the unique connection endpoint.

Q90

GATE-level

Coal India MT

TCP flow control prevents overwhelming the:

- (a) Network
- (b) Receiver
- (c) Sender
- (d) Router

Answer: (b)

Flow control (sliding window) protects the receiver; congestion control protects the network.

Q91

All 3 exams

Which TCP mechanism handles a fast sender and slow receiver?

- (a) Congestion control
- (b) Sliding window flow control
- (c) Routing
- (d) Fragmentation

Answer: (b)

The sliding window (receiver-advertised) implements flow control.

Q92

All 3 exams

Well-known ports occupy the range:

- (a) 0-1023
- (b) 1024-49151
- (c) 49152-65535
- (d) 0-255

Answer: (a)

Well-known ports = 0-1023; registered = 1024-49151; dynamic/private = 49152-65535.

B8. Application layer & services

Q93

All 3 exams

Which port does SMTP use?

- (a) 21
- (b) 23
- (c) 25
- (d) 110

Answer: (c)

SMTP uses TCP port 25 for sending mail.

Q94

All 3 exams

SSH uses which port?

- (a) 22
- (b) 23
- (c) 443
- (d) 80

Answer: (a)

SSH uses port 22 for encrypted remote login (Telnet=23, unencrypted).

Q95

All 3 exams

The DNS record that maps a name to an IPv4 address is:

- (a) MX
- (b) A
- (c) CNAME
- (d) PTR

Answer: (b)

An A record maps a hostname to an IPv4 address; AAAA maps to IPv6; MX to mail servers.

Q96

DSSSB TGT

All 3 exams

Which service allows remote file upload/download with separate control and data connections?

- (a) HTTP
- (b) FTP
- (c) SMTP
- (d) SNMP

Answer: (b)

FTP uses separate control (21) and data (20) channels for file transfer.

Q97

All 3 exams

HTTP is which type of protocol regarding state?

- (a) Stateful
- (b) Stateless
- (c) Connection-reserved
- (d) Circuit-switched

Answer: (b)

HTTP is stateless — each request is independent; cookies/sessions add state on top.

Q98

All 3 exams

Which protocol underlies web page transfer with encryption?

- (a) HTTP
- (b) HTTPS
- (c) FTP
- (d) Telnet

Answer: (b)

HTTPS = HTTP over TLS/SSL, encrypting web traffic (port 443).

Q99

All 3 exams

SSC/other

POP3 differs from IMAP in that POP3 typically:

- (a) Keeps mail synced on server
- (b) Downloads and removes mail from server
- (c) Sends mail
- (d) Encrypts mail

Answer: (b)

POP3 usually downloads messages to one device and removes them from the server; IMAP keeps them synced.

B9. IPv6, security & wireless

Q100

All 3 exams

A key advantage of IPv6 over IPv4 is:

- (a) Smaller header
- (b) Vastly larger address space
- (c) Requires NAT
- (d) Uses 32 bits

Answer: (b)

IPv6's 128-bit space is astronomically larger than IPv4's 32-bit, removing the need for NAT.

Q101

All 3 exams

Which is NOT a goal of network security (CIA triad)?

- (a) Confidentiality
- (b) Integrity
- (c) Availability
- (d) Compression

Answer: (d)

The CIA triad is Confidentiality, Integrity, Availability. Compression is not a security goal.

Q102

IBPS SO IT

All 3 exams

A man-in-the-middle attack primarily threatens:

- (a) Availability only
- (b) Confidentiality and integrity
- (c) Compression
- (d) Bandwidth

Answer: (b)

MITM lets an attacker read (confidentiality) and alter (integrity) communications.

Q103

All 3 exams

Which wireless standard family is commonly called Wi-Fi?

- (a) IEEE 802.3
- (b) IEEE 802.11
- (c) IEEE 802.15
- (d) IEEE 802.16

Answer: (b)

IEEE 802.11 = Wi-Fi. 802.3=Ethernet, 802.15=Bluetooth/PAN, 802.16=WiMAX.

Q104

All 3 exams

A proxy server primarily:

- (a) Assigns IPs
- (b) Acts as an intermediary between client and server
- (c) Routes between AS
- (d) Resolves names

Answer: (b)

A proxy sits between clients and servers, forwarding requests (for caching, filtering, anonymity).

Q105

DSSSB TGT

All 3 exams

VoIP technology is used for:

- (a) File storage
- (b) Voice calls over IP networks
- (c) Email
- (d) Web hosting

Answer: (b)

VoIP (Voice over IP) carries voice as data packets over IP networks (e.g. internet calling).

Q106

All 3 exams

Which device is needed to connect a home LAN to the internet via an ISP?

- (a) Repeater
- (b) Router (often with modem)
- (c) Hub
- (d) Bridge

Answer: (b)

A router (commonly combined with a modem) connects the home LAN to the ISP/internet.

Coverage & next

✓ What this bank covered

Section A (PYQ-style, ~50): exam-tagged questions modelled on real IBPS/SSC/GATE/PSU/DSSSB patterns across every CN sub-topic.

Section B (Practice, ~60): full syllabus-coverage revision set organised by sub-topic (B1-B9).

For IBPS (Aug 29): prioritise IBPS / All-3 tags; the GATE-tagged items (capacity math, ARP windows, subnet arithmetic, congestion phases) are Coal-India depth.

Same two-section format (PYQ + Practice) will follow for the other modules — OS and DBMS next per your IBPS sequence.